

GOLDEN GATE UNIVERSITY

Doctor of Business Administration (Generative AI)

Chapter 2: Literature Review

Complete Flow Document — ASD-Specific (Single-Disease Focal)

**A Governance Framework for Responsible Explainable
AI-Assisted Retrospective EEG-Based Neuropsychiatric
Decision Support Under Human Clinical Oversight**

*Integrating Generative AI, Retrieval-Augmented Generation,
and Continuous Monitoring for Clinical and Consumer Healthcare Applications*

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*This ASD-specific Ch. 2 flow document narrows the multi-disease companion
(Ch2_Flow_Document.pdf at 2,847 papers) to the single-disease ASD focal:
6 themes, 9 theories, PRISMA cascade, SWOT, CMM maturity, 5 hypotheses.*

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1. Chapter 2 Objective + Claim (ASD-Specific)

The Ch. 2 objective is to critically synthesise the ASD-EEG + AI-governance literature, identify research gaps, and ground 5 hypotheses in 9 theoretical foundations. The chapter claim is a single load-bearing sentence that examiners can quote in the viva.

Objective. *Systematically review the ASD-EEG + AI-governance literature (PRISMA-aligned 2,847 papers → 156 included), synthesise 6 thematic areas, identify 5 research gaps (G1–G5), and ground 5 hypotheses (H1–H5) in 9 theoretical foundations.*

Chapter claim. *“The existing ASD-EEG + AI-governance literature demonstrates high single-axis classification accuracy (78%–99%) but reveals five critical gaps (G1 theoretical + G2 practical + G3 technical + G4 governance + G5 methodological) that collectively prevent unified, explainable, and governance-embedded clinical AI deployment for pediatric ASD screening support under clinician oversight.”*

2. Process Flow / Journey Map

The Ch. 2 journey map is a structured walk through 7 stages from search strategy to chapter summary. Each stage produces a named artefact that examiners can grep against.

Section	Stage name	What it delivers
§2.1	Search strategy	PRISMA-aligned protocol; IEEE + PubMed + Scopus + WoS (2015–2025); 6 thematic codes.
§2.2	Conceptual foundations	Key constructs: clinical AI, multi-domain frameworks, explainability, agentic systems, governance, pediatric ASD.
§2.3	Nine theories	4 technical (Signal Detection + Computational Neuroscience + Radiomics + Transfer Learning) + 5 organisational (RBV + Dynamic Capabilities + TAM + Diffusion of Innovation + DSR).
§2.4	Six empirical themes	ASD-EEG biomarkers + classical ML + deep learning + XAI + governance + trust/adoption.
§2.5	Critical analysis + comparative matrix	Per-theme strengths / weaknesses / contradictions.
§2.6	Five research gaps G1–G5	Theoretical + practical + technical + governance + methodological.
§2.7	Five hypotheses H1–H5	Grounded in 9 theories; each falsifiable with decision criterion.
§2.8	Strategic analysis (SWOT + CMM)	Field maturity assessment; RGAIG positioning.
§2.9	Ethics + trust (RTAI)	10-dimension safeguard; clinician-in-the-loop oversight.
§2.10	Contradictions in literature	DL vs. ML debate; explainability-accuracy tension; single-jurisdiction validation.
§2.11	Chapter summary	Bridge to Ch. 3 methodology.

3. PRISMA Systematic-Review Cascade (ASD-Focused)

The PRISMA cascade narrows 2,847 retrieved papers down to 156 included papers across 6 thematic areas. ASD-EEG papers comprise the largest theme (45 / 156 = 29%).

Stage	Action	Output count	Criterion
A1	Database search (IEEE + PubMed + Scopus + WoS; 2015–2025)	2,847 papers	Search-term protocol with ASD-EEG + AI governance + XAI + HITL keywords.
A2	Title + abstract screening (PRISMA)	612 retained	Inclusion: ASD-EEG OR AI governance OR XAI OR HITL OR pediatric AI.
A3	Full-text review with quality assessment	156 included	Quality: peer-reviewed + reproducible methods + reported metrics.
B1	Code by ASD-relevant domain area	6 thematic areas	ASD-EEG biomarkers / classical ML / DL / XAI / governance / trust+adoption.
B2	Per-theme comparison tables	6 tables in main thesis	Per-method + per-accuracy + per-limitation.
B3	Theme size	ASD-EEG: 45 papers (largest)	Per-theme paper distribution to anchor empirical chapters.

4. Six Empirical Themes (ASD-Specific)

The 6 themes mirror v3 dissertation E38 with ASD specificity. Each theme has a position the dissertation defends.

T#	Theme	# papers	Dissertation position
T1	Neurophysiological basis (delta–gamma + connectivity + ERPs)	28	Biomarkers → pipeline feature extraction.
T2	Classical ML (SVM / RF / XGBoost)	22	70–90% accuracy baseline; RGAIG wraps with governance.
T3	Deep Learning (1D-CNN / 2D-CNN / EEGNet / Transformer)	35	82–94% accuracy; XAI bridges to clinical acceptance.
T4	Explainable + Interpretable AI	24	Two-layer (caregiver + clinician) is the contribution.
T5	AI Governance (NIST + ISO + EU AI Act + DPDP)	27	Integration across regimes is the unmet gap.
T6	Trust + Adoption (TAM / TOE / RBV / Trust / Reassurance)	20	Mediator-moderator model extends TAM.

5. Nine Theoretical Foundations

Per Ch. 2 §2.3, the dissertation grounds its contribution in 4 technical + 5 organisational theories; the table below names each + assigns its role.

#	Theory	Type	Role in dissertation
T1	Signal Detection Theory	Technical	EEG signal-vs-noise; H1 sensitivity / specificity.
T2	Computational Neuroscience	Technical	Frequency-band biomarkers; ERP markers.
T3	Radiomics	Technical	Feature-engineering paradigm; H3 features.
T4	Transfer Learning	Technical	KAU → ABIDE-II cross-cohort generalisation; H2.
T5	Resource-Based View (RBV)	Organisational	RGAIG maturity as VRIN organisational resource.
T6	Dynamic Capabilities	Organisation	Hospital absorptive + reconfiguration capability.
T7	Technology Acceptance Model (TAM)	Organisational	Adoption pathway; mediated by governance maturity.
T8	Diffusion of Innovation (DoI)	Organisation	Adopter categorisation; B2C primary + B2B-supporting.
T9	Design Science Research (DSR)	Organisational	DSR methodology for artefact-building contribution.

6. Five Research Gaps G1–G5

Per Ch. 2 §2.6, the literature has 5 typed gaps; RGAIG closes all 5.

Gap	Type	What is missing + RGAIG closure
G1	Theoretical	No integrated theory linking RAI + trust + adoption for pediatric AI screening. RGAIG: 5-layer construct.
G2	Practical	No deployable governance maturity instrument for healthcare AI. RGAIG: 6-level + 24-control catalogue.
G3	Technical	No agentic + RAG + HITL architecture with caregiver-readable XAI. RGAIG: 7-layer reference architecture.
G4	Governance	No multi-jurisdiction mapping (NIST + ISO + EU AI Act + DPDP). RGAIG: cross-walk.
G5	Methodological	DSR + PLS-SEM + qualitative-panel hybrid rarely used together. RGAIG: mixed-methods design.

7. Five Hypotheses H1–H5 (Falsifiable Decision Criteria)

Each hypothesis pairs with an explicit decision criterion + statistical test + threshold.

H	Hypothesis	Test	Decision criterion (falsifiability)
H1	Accuracy $\geq 85\%$ on ASD-EEG (RGAIG-wrapped)	10-fold CV + bootstrap	Mean accuracy ≥ 0.85 with 95% CI not crossing 0.85.
H2	DL outperforms classical ML for ASD-EEG	Paired t-test	$p < 0.05$ with Cohen's $d \geq 0.5$.
H3	Feature engineering contributes $\geq 5\%$ accuracy	Ablation + SHAP	Removing top- k features drops accuracy by $\geq 5\%$.
H4	Automation reduces $\geq 90\%$ manual effort	Timing comparison	Measured time saving $\geq 90\%$ of baseline.
H5	RAG explanations $\geq 80\%$ expert acceptance	Expert panel ($n=12$) Likert	$\geq 80\%$ rate explanation $\geq 4/5$ on plausibility.

8. SWOT + CMM Maturity Assessment

Per Ch. 2 §2.8, the biomedical-AI field is currently at CMM Level 1–2 (initial / repeatable); RGAIG lifts adopting organisations to Level 3–4 (defined / managed).

SWOT axis	Biomedical AI field (ASD focal) + RGAIG positioning
Strengths	High single-domain ASD-EEG accuracy (78–99%); KAU + ABIDE-II rich datasets; mature DL toolchain.
Weaknesses	Fragmented per-condition tools; opaque outputs; manual workflows; single-jurisdiction validation.
Opportunities	EU AI Act + ISO 42001 + NIST RMF + DPDP all live 2024–2026; rising governance demand; B2C caregiver primary segment under-served.
Threats	Regulatory delay; commoditisation; competing closed-vendor frameworks; clinical-trust erosion on first AI surprise.
CMM (field today)	Level 1–2 (initial / repeatable).
CMM (RGAIG enables)	Level 3 (defined) + Level 4 (managed) via 6-level maturity instrument.

9. Ethics + Trust (RTAI Principles)

Per Ch. 2 §2.9, 10 ethical dimensions + 3-axis trust framework (technical + clinical + organisational); collectively the Responsible + Trustworthy AI (RTAI) principles.

Dimension	Safeguard
Privacy / PII	De-identification + AES-256 + DPDP / GDPR / HIPAA.
Pediatric vulnerability	ICMR + DPDP pediatric safeguards; child assent + parental consent.
Consent / waiver	ICMR 2017 §5.5; prior IRB references for $n=131 + n=12$.
Fairness + bias	$DI \geq 0.80$; EO-gap $< 5\%$; per-subgroup audit weekly.
Hallucination	3-layer guard; residual $< 3\%$; Ragas faithfulness gate.
Overtrust risk	Plain-language uncertainty; SOS conceptual only.
AI disclosure	Appendix I AI Declaration; parallel Codex+Claude changelogs.
Trust (technical)	Accuracy + calibration + reliability.
Trust (clinical)	HITL gate + explanation + audit row visible.
Trust (organisational)	Procurement maturity instrument + multi-jurisdiction crosswalk.

10. Contradictions in the Literature

Per Ch. 2 §2.10, the dissertation identifies three named contradictions that the RGAIG framework resolves.

Contradiction	Resolution by RGAIG
DL vs. ML debate	RGAIG-agnostic: governance wraps either; choice driven by interpretability + cost trade-off, not technology dogma.
Explainability vs. accuracy	Two-layer XAI (caregiver + clinician) preserves DL accuracy while delivering plain-language layer.
Single-jurisdiction vs. universal	Multi-jurisdiction crosswalk (NIST + ISO + EU AI Act + DPDP) provides a universal-but-grounded framework.

11. Chapter Summary + Bridge to Ch. 3

The chapter summary names what Ch. 2 has established and how it feeds Ch. 3.

Chapter 2 summary. The ASD-EEG + AI-governance literature offers mature single-axis frameworks for biomarkers (delta–gamma + connectivity + ERPs), for classical ML (70–90% accuracy), for deep learning (82–94%), for explainability (SHAP / LIME / counterfactual / attention), for governance (NIST + ISO + EU AI Act + DPDP), and for trust + adoption (TAM + TOE + RBV + Trust + Reassurance). However, no prior work integrates these axes into a single, operationalisable, audit-bearing governance fabric for B2C-primary, clinician-supported pediatric ASD screening support. The proposed RGAIG framework closes this integration gap (G1–G5) through a 5-layer + transverse-audit architecture, anchored by 24 controls + 6-level maturity instrument + multi-jurisdiction crosswalk. **Bridge to Ch. 3:** the gap-closure design is operationalised via the pragmatist mixed-methods DSR design (Ch. 3 §3.4) that pairs PLS-SEM evidence with qualitative expert-panel triangulation.

End of ASD-specific Ch. 2 Flow Document. Companion to the multi-disease Ch1_Flow_Document.pdf + Ch2_Flow_Document.pdf; this version narrows the broader scope to single-disease ASD focal.